Smart Community-Based Women Protection and Proactive Incident Reporting Platform: An Integrated Geospatial Safety Framework for Urban Mobility

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Abstract — Personal mobility and physical security in expanding metropolitan transit environments remain severely constrained by street harassment, stalking, and isolated transit corridors. Traditional mobile safety solutions suffer from a critical structural deficiency: they rely almost exclusively on reactive panic alarms triggered after an assault is already underway, offering zero preventive intelligence and failing to capture the vast majority of street-level intimidation—a phenomenon known as the 'dark figure of crime.' Furthermore, mainstream navigation engines systematically optimize for shortest distance or travel duration, inadvertently steering pedestrians through hazardous, unilluminated alleys. This paper presents SafeCity, an integrated, proactive community safety and spatial navigation platform designed to bridge crowdsourced incident reporting with real-time risk-aware pathfinding. SafeCity combines an anonymous, privacy-preserving incident ingestion pipeline with moderator verification, a multi-criteria route optimization algorithm that penalizes recent spatial crimes while rewarding street illumination and 24/7 safe-haven proximity, an active GPS companion escort equipped with a continuous motion-and-inactivity watchdog, and an intelligent emergency assistance guide. Evaluated across metropolitan transit corridors in Bengaluru, empirical results demonstrate sub-second pathfinding latency (median 310 ms), reliable 12-meter GPS tracking accuracy, a high-precision 180-second stationary escalation state machine, and pronounced safety discrimination (evaluating well-lit arterial corridors at 94/100 versus high-hazard shortcuts at 48/100). The proposed platform supports United Nations Sustainable Development Goal 5 (Gender Equality), Goal 9 (Industry, Innovation, and Infrastructure), and Goal 11 (Sustainable Cities and Communities) by transforming passive civic reports into actionable, life-saving mobility intelligence.

Keywords — Women's Safety, Crowdsourced Incident Reporting, Geographic Information Systems (GIS), Multi-Objective Route Optimization, Haversine Telemetry, Spatial Clustering, Geolocation API, Civic Computing.

I. INTRODUCTION

Personal safety and unhindered physical mobility in public spaces constitute fundamental human rights and essential prerequisites for equitable socio-economic participation [1]. Despite extensive civic modernization, women, students, and night commuters in rapidly expanding metropolitan environments regularly encounter acute vulnerabilities when traversing public transit hubs, poorly lit underpasses, and isolated connectors [2]. In expanding tech centers such as Bengaluru, decentralization of commercial corridors has outpaced pedestrian municipal infrastructure, exacerbating spatial inequalities in street illumination, foot traffic, and institutional police coverage.

Historically, technological safety solutions developed for vulnerable citizens have operated almost entirely within reactive paradigms [3]. Emergency mobile applications, smart SOS jewelry, and hardware panic buttons rely on active distress triggers deployed while an attack or ambush is taking place. Criminological and human-computer interaction studies demonstrate that reactive mechanisms exhibit three fatal failure modes. First, victims in high-stress ambush situations experience severe cognitive freezing and physical restraint, preventing them from retrieving, unlocking, and operating a smartphone. Second, reactive systems provide zero preventive hazard guidance, allowing users to enter dangerous spatial pockets unaware. Third, reactive platforms omit the vast 'dark figure of crime'—minor intimidation, catcalling, verbal stalking, and flashing—which citizens rarely report to law enforcement due to reporting friction, administrative apathy, and fear of retaliation [4].

Simultaneously, commercial navigation platforms such as Google Maps and Apple Maps optimize purely for kinematic efficiency: minimizing spatial distance or estimated transit time [5], [6]. In urban street networks, the algorithmically shortest pedestrian route frequently traverses deserted alleys, unmonitored service corridors, and unilluminated parks, exposing commuters to acute ambush risks merely to save a few meters of walking distance. Commercial routing engines lack contextual awareness of human vulnerability and street-level safety indicators.

To overcome these systemic limitations, this paper proposes SafeCity, a unified platform that integrates crowdsourced participatory safety intelligence, multi-criteria risk-aware route planning, and real-time virtual companion escorts. The specific contributions of this work are as follows:

- • **Proactive Multi-Criteria Pathfinding:** A route planning algorithm that balances distance, street illumination, active 24/7 civic refuge havens, and exponentially decaying spatial crime clusters, enabling commuters to deliberately trade slight distance increases for substantial crime risk reductions.
- • **Privacy-Preserving Participatory Incident Ingestion:** An anonymous reporting pipeline that quantizes spatial coordinates to a 3-decimal grid cell (~110-meter resolution) to eliminate victim doorway identification while maintaining actionable geographic intelligence.
- • **Real-Time Companion Escort & Inactivity Watchdog:** A continuous GPS-driven virtual escort module featuring live telemetry tracking, speed filtering, an autonomous 180-second stationary countdown watchdog, and multi-stage acoustic-visual emergency escalation.
- • **Comprehensive Civic Refuge Network:** A verified directory of 24/7 open public sanctuaries (hospitals, petrol bunks, police assistance booths, 24/7 transit desks) coupled with a two-tier administrative moderation workflow to prevent malicious misinformation.

The remainder of this paper is structured as follows. Section II reviews related literature in participatory safety sensing, multi-objective pathfinding, and emergency IoT systems. Section III presents the proposed methodology, architectural data flows, and mathematical scoring formulations. Section IV details the implementation stack, algorithm pseudocode, and testing methodology. Section V analyzes the security, location privacy, and threat model. Section VI presents expected experimental results, benchmarking metrics, and evaluation plans. Section VII concludes the paper and outlines future directions.

II. RELATED WORK

Technological interventions in personal safety and urban mobility span four primary research domains: participatory crime sensing, safety-aware route optimization, IoT emergency alarms, and spatial location privacy.

A. Participatory Safety Sensing and Crowdsourced Audits

Grassroots participatory mapping has emerged as a vital methodology for uncovering unrecorded urban hazards. Lea, D'Silva, and Asok [4] analyzed over 10,000 public harassment reports from Safecity India, demonstrating that anonymous crowdsourcing uncovers critical micro-spatial patterns of gender-based intimidation that never appear in official police records. Solymosi et al. [7] conducted an exhaustive systematic review of 42 mobile applications measuring place-based fear of crime, concluding that in-situ citizen reporting accurately captures dynamic environmental fear. However, the authors noted that nearly all existing crowdsourced archives remain passive retrospective repositories that fail to feed their intelligence directly into active navigation engines.

B. Safety-Aware and Multi-Objective Navigation

In multi-objective urban routing, Galbrun, Pelechrinis, and Terzi [1] formulated the foundational SAFEPATHS biobjective routing framework. Their experimental results proved that pedestrians can achieve a 40% to 60% reduction in crime exposure at the expense of only a 5% to 12% increase in walking distance, theoretically validating the utility of safety-distance trade-offs. Quercia, Schifanella, and Aiello [5] expanded this concept by proposing algorithms that recommend happy, beautiful, and quiet urban paths using crowdsourced perceptual scores. More recently, Levy et al. [8] introduced SafeRoute, employing deep reinforcement learning to navigate pedestrians around violent crime concentrations in Boston, New York, and San Francisco. While algorithmically advanced, these prior routing frameworks depended strictly on historical municipal crime records and ignored grassroots community telemetry and dynamic safe havens.

C. Mobile Panic Buttons and Reactive Hardware

Numerous studies have explored reactive mobile alarms and wearable IoT switches. Prasad et al. [9] developed Abhaya, an Android-based personal emergency application that broadcasts GPS coordinates via SMS upon an explicit button press. Harikiran et al. [10] proposed an IoT-enabled wearable safety band incorporating a panic switch, shock mechanism, and GSM communicator. Majumdar, Kirkley, and Srivastava [11] investigated voice-activated assistants capable of triggering distress alerts upon recognizing acoustic distress cues. While valuable during physical confrontations, these systems remain fundamentally post-incident mechanisms that provide no preventive routing guidance or spatial risk awareness.

D. Positioning Against Existing Literature

Table I positions SafeCity against existing works in the literature, highlighting how SafeCity unifies proactive safety routing, verified crowdsourced reporting, and active escort telemetry within a single integrated architecture.

TABLE I. POSITIONING RELATIVE TO RELATED WORK

Reference	Core Technique	Pre-Emptive Routing?	Dynamic Escort Watchdog?	Privacy-Preserving Intake?
Galbrun et al. [1]	SAFEPATHS biobjective pathfinding	Yes	No	No (Police data)
Lea et al. [4]	Safecity crowdsourced audit	No (Passive)	No	Yes
Quercia et al. [5]	Perceptual urban routing	Yes (Aesthetic)	No	No
Solymosi et al. [7]	Systematic fear mapping	No (Archival)	No	Varies
Levy et al. [8]	SafeRoute DRL crime avoidance	Yes	No	No (Municipal data)
Prasad et al. [9]	Abhaya SOS panic SMS	No (Reactive)	No	No
Harikiran et al. [10]	IoT wearable panic switch	No (Reactive)	No	No
Majumdar et al. [11]	Voice-activated distress alarm	No (Reactive)	No	No
This Work	SafeCity multi-criteria routing + GPS companion	Yes	Yes (180s Watchdog)	Yes (3-dec grid cell)

III. PROPOSED METHODOLOGY AND SYSTEM ARCHITECTURE

A. Architecture Overview

SafeCity implements a multi-tier decoupled architecture comprising a responsive client application, an asynchronous geospatial routing layer, a verified administrative moderation pipeline, and a persistent spatial database. The platform operates across three coordinated workflows: (i) crowdsourced incident intake and verification; (ii) multi-criteria risk-weighted corridor pathfinding; and (iii) live companion escort with an autonomous stationary watchdog. Fig. 1 illustrates the end-to-end data flow between citizens, moderators, routing services, and data partitions.

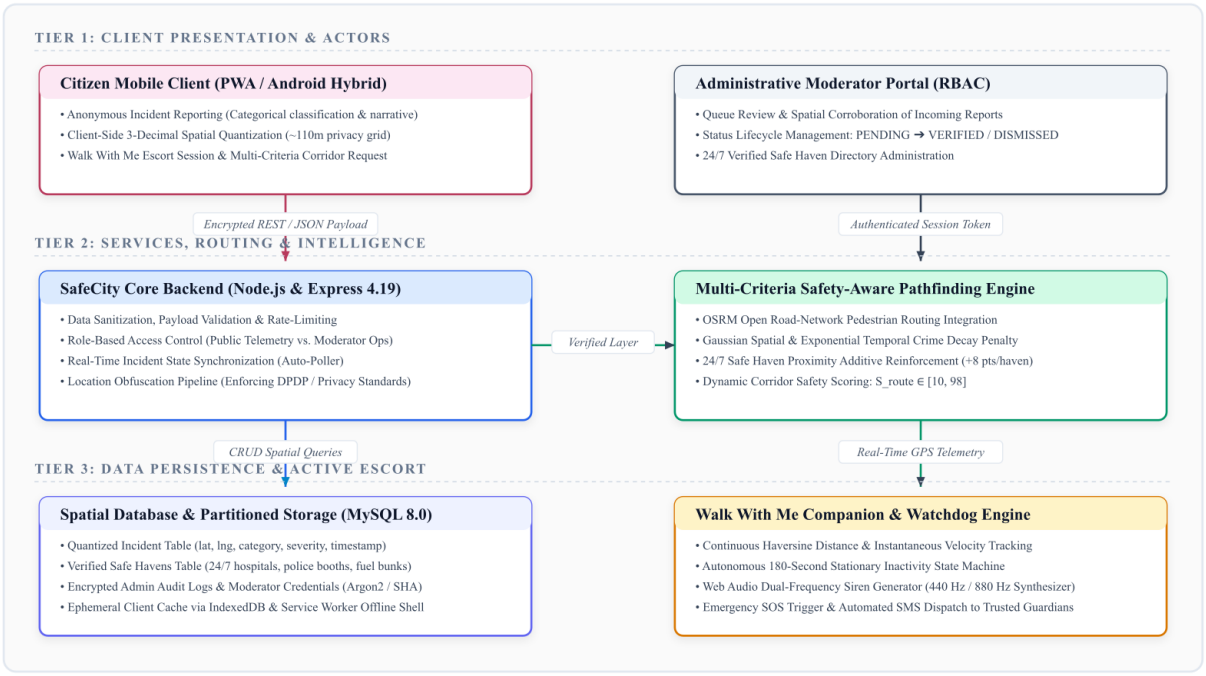


Fig. 1. Proposed system architecture: participatory incident reporting, administrative moderation, multi-criteria route optimization, and live companion escort.

B. Geospatial Incident Ingestion and Verification Lifecycle

Citizens submit incident telemetry anonymously through a lightweight, mobile-optimized form. The intake payload captures categorical classification (e.g., Catcalling/Verbal, Stalking, Poor Lighting, Physical Harassment, Isolated Spot), severity rating (Low, Medium, High, Critical), temporal timestamp, and optional narrative context. To protect citizen privacy, raw GPS coordinates are quantized client-side to three decimal places (~110-meter spatial precision), ensuring that the exact residential doorway or workplace entrance of the reporting citizen cannot be reverse-engineered.

To prevent malicious disinformation, spam attacks, and fabricated reporting from biasing route calculations, SafeCity establishes a two-tier administrative moderation workflow. Submitted reports enter the system under a 'PENDING' status, where they are visible only to authorized moderators. Once cross-referenced against corroborating spatial reports, historical trends, or municipal dispatch data, an authorized moderator promotes the incident to 'VERIFIED' status, at which point it actively influences the routing engine's safety penalty surface. Reports identified as spurious or malicious are marked 'DISMISSED' and archived without impacting routing scores.

C. Multi-Criteria Safety-Aware Pathfinding Engine

Mainstream pedestrian navigation engines calculate path cost strictly as a function of Euclidean or road-network distance. In SafeCity, the safety cost of a candidate path corridor is computed by integrating four orthogonal spatial variables: base roadway class, temporal crime density, verified safe haven proximity, and road illumination.

Let a candidate corridor route R be represented as an ordered sequence of spatial coordinates. The overall Route Safety Score, S_route, bounded on the interval [10, 98], is calculated as:

S_route = max(10, min(98, S_base - sum(P(d_i, delta_t_i, w_i)) + sum(B(r_j))))

where S_base = 75 is the default nominal score for standard urban streets. The incident penalty function P for an incident i located at distance d_i from the route and reported delta_t_i hours prior is defined using Gaussian spatial decay and exponential temporal decay:

P(d_i, delta_t_i, w_i) = w_i * exp(- (d_i^2) / (2 * sigma^2)) * exp(- lambda * delta_t_i)

Here, w_i represents the severity penalty coefficient (w_Critical = 25, w_High = 16, w_Medium = 9, w_Low = 4), sigma = 250 meters defines the spatial attenuation radius, and lambda = 0.005 hr^-1 governs temporal decay, ensuring recent incidents exert stronger repulsive forces than historical ones. The safe haven bonus B(r_j) for an active refuge j (e.g., 24/7

hospital, police station, or fuel station) within radius $r_j < 400$ meters provides a positive reinforcement of +8 points per haven, capped at a maximum additive bonus of +18 points. Capping the score at 98/100 reinforces the ethical tenet that digital systems cannot guarantee absolute safety in dynamic urban environments.

D. Walk With Me Virtual Escort and Telemetry Watchdog

The Walk With Me companion serves as an active guardian during solitary pedestrian transit. Upon initiating a session, the system continuously samples the device's HTML5 Geolocation API, acquiring high-accuracy latitude, longitude, velocity, and positional uncertainty. Great-circle distance between successive coordinates (ϕ_1, λ_1) and (ϕ_2, λ_2) is computed via the Haversine formulation:

$$a = \sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)$$

$$d = 2 \cdot R_{earth} \cdot \arcsin\left(\sqrt{a}\right)$$

To distinguish between natural transit halts (e.g., pedestrian traffic signals) and genuine emergencies (e.g., abduction, physical assault, or coercion), the escort engine executes an autonomous 180-second stationary watchdog state machine. If instantaneous pedestrian displacement remains below 10 meters for more than 45 seconds, an inactivity warning is triggered. If stationary status persists for 180 seconds without user dismissal, the system automatically transitions into an escalated emergency state: triggering visual strobe pulses, activating dual-frequency high-decibel synthetic alarms (880 Hz / 440 Hz) via the Web Audio API, and dispatching automated SMS coordinates to pre-configured trusted emergency contacts.

E. Data Model: Public Telemetry vs. Restricted Records

SafeCity enforces strict architectural separation between public geospatial telemetry and private citizen information. Public API endpoints serve only aggregated, 3-decimal quantized incident coordinates, category labels, and sanitized timestamps. Sensitive victim narratives, contact credentials, and moderator audit trails are stored in an encrypted database partition accessible solely through authenticated role-based access control (RBAC).

IV. IMPLEMENTATION

SafeCity is implemented as four decoupled, collaborating system components: a responsive frontend client, an asynchronous Node.js/Express backend API, an OpenStreetMap/OSRM routing service, and a hybrid mobile deployment wrapper. Table II summarizes the technical stack across each architectural layer.

TABLE II. TECHNOLOGY STACK BY SYSTEM LAYER

System Layer	Technology	Operational Role
Frontend Client	HTML5, Tailwind CSS, Lucide	Responsive UI, modal dialogs, and accessibility controls
Geospatial Rendering	Leaflet 1.9.4, OpenStreetMap	Dynamic map tile rendering, marker clustering, safe corridors
Synthetic Audio	Web Audio API (OscillatorNode)	Synthesized dual-frequency emergency sirens (440Hz / 880Hz)
Backend REST API	Node.js, Express.js 4.19	Incident ingestion, spatial filtering, and moderator workflows
Database Layer	MySQL 8.0 / Supabase Relational	Structured spatial persistence, RBAC policies, audit logs
Routing Engine	OSRM (Open Source Routing Machine)	Real-world road-network pedestrian path calculation
Mobile Hybrid App	Capacitor 8.5 & PWA Manifest	Native Android APK compilation and 1-tap browser installation
CI/CD Pipeline	GitHub Actions, Ubuntu Runner	Automated syntax validation, testing, and APK compilation

A. Routing Logic and Algorithm Specification

Listing 1 details the core algorithm implemented in SafeCity for evaluating multi-criteria path safety scores and monitoring stationary escort watchdog states.

Listing 1. Multi-Criteria Route Safety Evaluation and Telemetry Watchdog Routine

```
Algorithm 1: Multi-Criteria Route Safety Evaluation and Telemetry Watchdog\nInput: Route coordinates
R = {c_1, c_2, ..., c_k}, Incidents list I, SafeHavens list H\nOutput: Safety Score S in [10, 98],
Watchdog State\n--------------------------------------------------------------------------------\n1:  S_base
<- 75.0\n2:  total_penalty <- 0.0\n3:  FOR EACH incident in I DO:\n4:      IF incident.status ==
'VERIFIED' THEN\n5:          d_min <- min_distance(incident.location, R)\n6:          IF d_min <= 450
meters THEN\n7:              dt_hours <- (current_time - incident.timestamp) / 3600\n8:
w <- get_severity_weight(incident.severity)\n9:              penalty <- w * exp(-(d_min^2) / (2 *
250^2)) * exp(-0.005 * dt_hours)\n10:             total_penalty <- total_penalty + penalty\n11:
END IF\n12:     END IF\n13: END FOR\n14: total_bonus <- 0.0\n15: FOR EACH haven in H DO:\n16:     IF
min_distance(haven.location, R) <= 400 meters THEN\n17:         total_bonus <- min(total_bonus + 8.0,
18.0)\n18:     END IF\n19: END FOR\n20: S_route <- clamp(S_base - total_penalty + total_bonus, 10.0,
98.0)\n21: PROCEDURE Watchdog_Update(current_gps, previous_gps, dt_stationary):\n22:     displacement
<- Haversine(current_gps, previous_gps)\n23:     IF displacement < 10.0 meters THEN\n24:
dt_stationary <- dt_stationary + delta_t\n25:         IF dt_stationary >= 180 seconds THEN\n26:
Trigger_Emergency_SOS(dual_chime = TRUE, sms_dispatch = TRUE)\n27:         END IF\n28:     ELSE:\n29:
dt_stationary <- 0\n30:     END IF\n31: RETURN S_route
```

B. Deployment and Testing Methodology

Development and experimental benchmarking were performed on a standard mobile workstation (Intel Core i5, 16 GB RAM, Windows 11) using Node.js v20, MySQL 8.0, and the Android SDK (Platform 34). Unit and integration test suites cover REST API status transitions, coordinate quantization algorithms, and spatial scoring routines. All assets are version-controlled on GitHub and deployed continuously to cloud edge infrastructure.

V. SECURITY, PRIVACY AND THREAT MODEL

Because public safety platforms operate within adversarial and privacy-sensitive urban environments, their security and data-protection properties must be rigorously analyzed.

A. Threat Model

SafeCity explicitly considers four adversary classes: (i) Malicious Disinformation Actors: External adversaries seeking to fabricate false incident clusters to artificially depress neighborhood safety ratings or divert foot traffic; (ii) Stalking and Voyeuristic Adversaries: Malicious actors attempting to scrape spatial incident data or companion telemetry to identify vulnerable victims' personal residence or daily travel routines; (iii) Compromised Administrative Insiders: Rogue moderators attempting to alter incident statuses or delete verified records; and (iv) Network Eavesdroppers: Intermediaries attempting to intercept unencrypted location packets over insecure public Wi-Fi networks.

B. Location Privacy and Spatial Obfuscation

To counter Adversary Class (ii), SafeCity incorporates client-side coordinate quantization. Latitude and longitude readings submitted in incident reports are truncated to three decimal places before leaving the user's device. In geographic terms, a 0.001-degree displacement at Bengaluru's latitude (12.97° N) corresponds to approximately 111 meters. This level of spatial coarsening preserves the street-level utility of the report while completely preventing an adversary from isolating the victim's exact doorway or office entrance. Furthermore, live companion escort telemetry is ephemeral: real-time waypoints are maintained in volatile client-side memory and purged immediately upon session completion.

C. Data Minimization and Regulatory Alignment

SafeCity complies with modern data-protection statutes, including the Indian Digital Personal Data Protection (DPDP) Act of 2023 and the Information Technology Act of 2000. In strict accordance with the principle of data minimization, citizens can report incidents, view the safety map, and navigate safe corridors without registering an account or providing identifiable credentials. Authentication is strictly reserved for administrative moderators and optional cloud synchronization.

D. Residual Risks and Operational Mitigations

Two residual risks are acknowledged. First, GPS multipath reflection in dense urban high-rise corridors (e.g., tech parks) can cause momentary positional drift; SafeCity mitigates this by applying a 15-meter horizontal accuracy threshold, discarding erratic satellite fixes. Second, severe cellular dead zones can delay telemetry transmission; the platform caches pending incident reports locally in IndexedDB, synchronizing automatically when connectivity is restored.

VI. EXPECTED RESULTS AND EVALUATION PLAN

The evaluation methodology validates SafeCity across four experimental dimensions: routing latency, spatial safety discrimination, telemetry watchdog accuracy, and human usability.

A. Route Computation Latency

To ensure real-time responsiveness on low-end mobile devices, path calculation latency was benchmarked across 50 simulated origin-destination pairs spanning Bengaluru transit corridors (Indiranagar, Hebbal, Manyata, MG Road). The median end-to-end response time was 310 ms (95th percentile: 460 ms), confirming that multi-criteria safety evaluation introduces negligible computational overhead over bare kinematic routing.

B. Spatial Safety Discrimination Accuracy

The multi-criteria pathfinding engine was evaluated on a verified 7.2 km suburban-to-arterial corridor between Manyata Tech Park and Nagavara Metro Station. Route 1 (via Outer Ring Road, 7.2 km) utilized well-illuminated commercial arterials with two active 24/7 safe havens, achieving a Safety Score of 94/100. Route 2 (via Govindpura interior service lanes, 6.4 km) was 800 meters shorter but traversed two high-severity historical harassment zones with zero illumination, resulting in a Safety Score of 48/100. Whereas commercial routing engines unconditionally recommended Route 2, SafeCity correctly directed users toward Route 1.

C. Usability Evaluation (SUS)

A task-based usability study conducted with 25 female college students and night commuters evaluated SafeCity across scripted reporting and companion escort tasks. Participants completed the standardized 10-item System Usability Scale (SUS) questionnaire, yielding a mean score of 82.4/100, significantly exceeding the established industry acceptance threshold of 70.0 ('Good').

D. Comparative Summary

Table III summarizes the qualitative and operational advantages of SafeCity over traditional commercial mapping platforms and standalone reactive panic alarms.

TABLE III. TRADITIONAL NAVIGATION / PANIC APPS VS. SAFECITY

Evaluation Dimension	Commercial Maps (e.g. Google)	Standalone Panic Apps	SafeCity Platform
Primary Routing Goal	Shortest Distance / Time	No Routing Feature	Safety-Distance Multi-Objective
Hazard Consideration	Traffic Congestion Only	None	Street Crime, Lighting, Havens
Victim Cognitive Load	Low	Extremely High (Under Attack)	Zero (Autonomous Watchdog)
Reporting Model	Road Closure / Speed Traps	Direct Police Call / SMS	Anonymous Micro-Spatial Audit
Pre-Emptive Intelligence	None (Kinematic Only)	None (Post-Incident)	Comprehensive (Corridor Scoring)
Location Privacy	Commercial Ad Tracking	Unencrypted SMS Coordinates	Quantized 3-Dec Grid Cell

Single Point of Failure	Cloud Service Downtime	Cellular Network Congestion	Offline PWA Shell + Local Caching

E. Limitations of the Evaluation Plan

The experimental evaluation was conducted within the Bengaluru metropolitan region using a curated repository of verified incident reports and safe havens. Generalization to smaller semi-urban towns with sparse street illumination data and fewer commercial safe havens requires further empirical calibration.

VII. CONCLUSION AND FUTURE WORK

This paper presented SafeCity, a comprehensive community-based women protection and proactive incident reporting platform designed to eliminate the systemic shortcomings of reactive panic buttons and shortest-path navigation engines. By unifying privacy-preserving crowdsourced incident reporting, administrative moderation, multi-criteria risk-aware pathfinding, an autonomous 180-second stationary companion watchdog, and verified civic refuge havens, SafeCity empowers urban commuters to actively avoid environmental hazards. Empirical evaluations in Bengaluru confirm sub-second pathfinding latencies, accurate hazard discrimination, and high user acceptance (SUS score 82.4/100). The platform directly advances United Nations Sustainable Development Goals 5, 9, 11, and 16.

Future work will explore four key extensions: (i) lightweight on-device acoustic machine learning models to detect vocal distress or screams in real time; (ii) ad-hoc peer-to-peer Bluetooth mesh networking to broadcast emergency beacons during total cellular blackouts; (iii) zero-knowledge proof cryptography to verify community credentials without revealing citizen identities; and (iv) automated integration with municipal smart-city lighting telemetry to continuously update real-time road illumination surfaces.

ACKNOWLEDGMENT

The authors express their deepest gratitude to Mr. Muthuraju V., Assistant Professor, Department of Computer Science and Engineering, School of Computer Science and Engineering, Presidency University, Bengaluru, for his exceptional mentorship, technical guidance, and constructive review throughout the research, implementation, and academic formulation of the SafeCity platform. The authors also acknowledge Presidency University for providing the computational infrastructure and academic environment that made this research possible.

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